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Q1. Photosynthesis is the process by which green plants use sunlight, absorbed by chlorophyll, to make glucose from carbon dioxide and water, releasing oxygen.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Q2. Ohm's law: at constant temperature, the current through a conductor is directly proportional to the potential difference across it, V = IR.

R = 12 V / 3 A = 4 Ω

Q3. Acids turn blue litmus red and have a pH below 7; bases turn red litmus blue and have a pH above 7.

Neutralisation is the reaction of an acid with a base to form a salt and water, for example $HCl + NaOH \rightarrow NaCl + H_2O$.